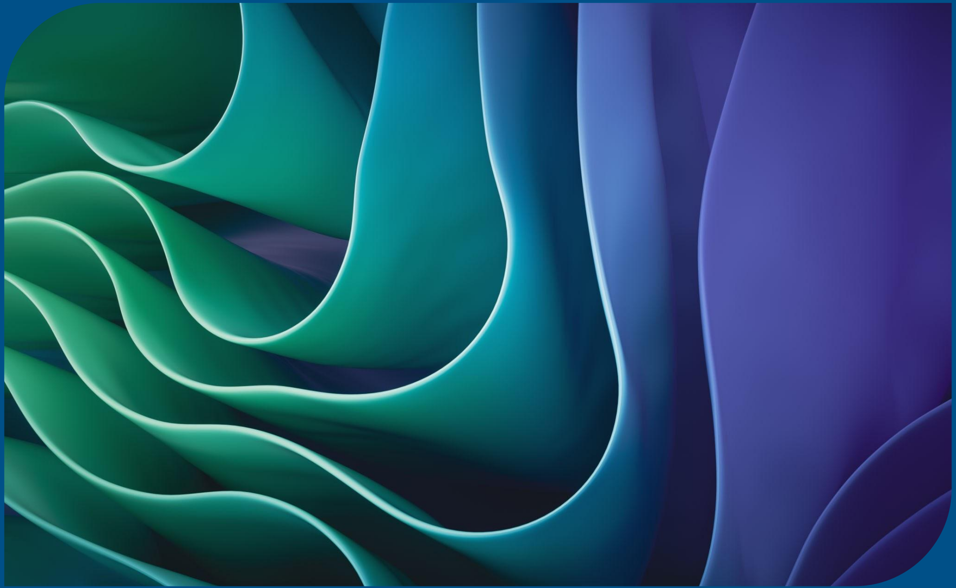
Philosophy, Trends and the Big Picture

IS2205- Mobile Application Design and Development



1. Ubiquitous and Ambient Computing
2. Application Distribution
3. Network Technology, Security and Privacy
4. Progressive and Cross-platform Apps
5. Other Topics


INTEGRATED USER ECOSYSTEM: CONNECTED DEVICES
SMART HOME HUB:
Controls & Automation
22°C
ON
SECURE
AR GLASSES:
Heads-Up Display
Arrow
3.5 mi
SMART HOME HUB:
Controls & Automation
AR GLASSES:
Heads-Up Display
78 BPM
8420
New MSG:
John
MAP
SMARTWATCH:
Health & Notifications
CONNECTED CAR:
In-Vehicle Infotainment

Ubiquitous and Ambient Computing

Computing is everywhere, embedded in the environment rather than tied to a single device – Mark Weiser of Xerox

Software that reacts to human presence and intent without needing explicit app launches

Ubiquitous and Ambient Computing

Ubiquitous computing names the third wave in computing, just now beginning. First were mainframes, each shared by lots of people. Now we are in the personal computing era, person and machine staring uneasily at each other across the desktop. Next comes ubiquitous computing, or the age of calm technology, when technology recedes into the background of our lives.

—Mark Weiser

Discussion Question:

When smart glasses or ambient voice interfaces replace the phone screen, what happens to the concept of an 'App'?

Are we developing software for displays, or are we developing software for human intent?

From network to self-healing infrastructure and responsible innovation

Rise of AI-native networks
Self-healing infrastructure
Multicloud and platform engineering
Ethical innovation ?
Sustainable telecom engineering ?
From telecom as a vertical to a universal horizontal

https://www.hcltech.com/en-us/trends-and-insights/telecom-trends-2026

5G Networks

Fast data transmission: 5G speeds can reach up to 10 Gbps

Ultra-low latency: Reduced to just 1 millisecond, making mobile gaming, video streaming, and AR/VR experiences smoother

Enhanced IoT capabilities: Supporting 1 million devices per square kilometer

The App Store Friction

Installation barriers

Binary size constraints

Regulatory/antitrust shifts

Update latency

Progressive Web Apps (PWAs)

WebAssembly (Wasm) : Allows portable C++/Rust code to be run on a browser environment

The line between "Web" and "Native" is blurring

Since C++/Rust engines run directly inside browser can be close to native speed

Discussion Question

If Progressive Web Apps (PWAs) running on WebAssembly can match native app performance without requiring an App Store download or a 30% revenue fee, why do native App Stores still dominate?

Is it a technical limitation or a business monopoly?

Cross-Platform Evolution

Generation 1: WebViews (Cordova/Ionic) → High friction, slow rendering.

Generation 2: Bridges (React Native) → Native controls tied to async bridges.

Generation 3: Direct Canvas Engine (Flutter) → Compiled native binaries rendering pixels directly.

Generation 4: Unified Language Targets (Kotlin Multiplatform) → Shared business logic, native platform UI.

Discussion Question

There are multiple cross-platform frameworks in use. In the future what changes will come to these frameworks and their specific native platforms?

Security and Privacy Aspect

Biometric authentication: Apps are integrating fingerprint, facial recognition, and voice authentication

Zero-trust security models: Instead of assuming trust once a user logs in, apps continuously verify identity, monitor access patterns, and detect anomalies to prevent unauthorized access

GDPR & CCPA compliance: With global regulations tightening, apps must follow stricter data privacy laws, ensuring users are suppose to have more control over their personal information and how it’s stored or shared

GDPR & CCPA : To Read

An extract is available in the VLE

https://www.okta.com/blog/industry-insights/ccpa-vs-gdpr/

Ethical Aspects of Mobile Apps

Implementing Ethics for a Mobile App Deployment (2016) : To Read

Abstract
Supporting Autonomy
Withdrawal
Privacy Risks
Uncertainty in ethics

https://johnrooksby.org/papers/OzCHI16-rooksby-implementingEthics.pdf

What about the industry?

Uncertainty

Politics

Resilience

Non-technical skills

Technical skills

Work-environment

The End